John Doe

Machine learning engineer · efficient models and reliable infrastructure

San Francisco, CA | john.doe@email.com | rendercv.com | rendercv | rendercv

Summary

I develop efficient machine learning systems, from model architecture and inference kernels to dependable training infrastructure. My work combines applied research with tools that help engineering teams bring promising ideas into production.

Experience

Co-Founder & CTO, Nexus AI
June 2023 – present
3 years 5 months
San Francisco, CA

- Built foundation model infrastructure serving 2M+ monthly API requests with 99.97% uptime
- Raised $18M Series A led by Sequoia Capital, with participation from a16z and Founders Fund
- Scaled engineering team from 3 to 28 across ML research, platform, and applied AI divisions
- Developed proprietary inference optimization reducing latency by 73% compared to baseline

Research Intern, NVIDIA Research
May 2022 – Aug 2022
4 months
Santa Clara, CA

- Designed sparse attention mechanism reducing transformer memory footprint by 4.2x
- Co-authored paper accepted at NeurIPS 2022 (spotlight presentation, top 5% of submissions)

Research Intern, Google DeepMind
May 2021 – Aug 2021
4 months
London, UK

- Developed reinforcement learning algorithms for multi-agent coordination
- Published research at top-tier venues with significant academic impact
- ICML 2022 main conference paper, cited 340+ times within two years
- NeurIPS 2022 workshop paper on emergent communication protocols
- Invited journal extension in JMLR (2023)

Research Intern, Apple ML Research
May 2020 – Aug 2020
4 months
Cupertino, CA

- Created on-device neural network compression pipeline deployed across 50M+ devices
- Filed 2 patents on efficient model quantization techniques for edge inference

Education

PhD, Computer Science
Princeton University
Sept 2018 – May 2023
Princeton, NJ

- Thesis: Efficient Neural Architecture Search for Resource-Constrained Deployment
- Advisor: Prof. Sanjeev Arora
- NSF Graduate Research Fellowship, Siebel Scholar (Class of 2022)

BS, Computer Engineering
Boğaziçi University
Sept 2014 – June 2018
Istanbul, Türkiye

- GPA: 3.97/4.00, Valedictorian
- Fulbright Scholarship recipient for Graduate Studies

Skills

Languages: Python, C++, CUDA, Rust, Julia, SQL, Bash

ML Frameworks: PyTorch, JAX, TensorFlow, Triton, ONNX, TensorRT, Hugging Face, scikit-learn

Infrastructure: Kubernetes, Ray, Distributed training, GPU profiling, Model serving, Experiment tracking, AWS, GCP

Research Intern, Microsoft Research
May 2019 – Aug 2019
4 months
Redmond, WA

- Implemented novel self-supervised learning framework for low-resource language modeling
- Research integrated into Azure Cognitive Services, reducing training data requirements by 60%

Research Areas: Neural architecture search, Model compression, Efficient inference, Sparse attention, Mixture of experts, Multi-agent learning, Reproducible evaluation

Projects

FlashInfer

Open-source library for high-performance LLM inference kernels, Jan 2023 – present

- Achieved 2.8x speedup over baseline attention implementations on A100 GPUs
- Adopted by 3 major AI labs, 8,500+ GitHub stars, 200+ contributors

NeuralPrune

Automated neural network pruning toolkit with differentiable masks, Jan 2021

- Reduced model size by 90% with less than 1% accuracy degradation on ImageNet
- Featured in PyTorch ecosystem tools, 4,200+ GitHub stars

Publications

Sparse Mixture-of-Experts at Scale: Efficient Routing for Trillion-Parameter Models
July 2023
John Doe, Sarah Williams, David Park
10.1234/neurips.2023.1234 (NeurIPS 2023)

Neural Architecture Search via Differentiable Pruning
Dec 2022
James Liu, *John Doe*
10.1234/neurips.2022.5678 (NeurIPS 2022, Spotlight)

Multi-Agent Reinforcement Learning with Emergent Communication
July 2022
Maria Garcia, *John Doe*, Tom Anderson
10.1234/icml.2022.9012 (ICML 2022)

On-Device Model Compression via Learned Quantization
May 2021
John Doe, Kevin Wu
10.1234/iclr.2021.3456 (ICLR 2021, Best Paper Award)

Selected Honors

- MIT Technology Review 35 Under 35 Innovators (2024)
- Forbes 30 Under 30 in Enterprise Technology (2024)
- ACM Doctoral Dissertation Award Honorable Mention (2023)
- Google PhD Fellowship in Machine Learning (2020 – 2023)
- Fulbright Scholarship for Graduate Studies (2018)

Patents

1. Adaptive Quantization for Neural Network Inference on Edge Devices (US Patent 11,234,567)
2. Dynamic Sparsity Patterns for Efficient Transformer Attention (US Patent 11,345,678)
3. Hardware-Aware Neural Architecture Search Method (US Patent 11,456,789)

Invited Talks

4. Scaling Laws for Efficient Inference — Stanford HAI Symposium (2024)
3. Building AI Infrastructure for the Next Decade — TechCrunch Disrupt (2024)
2. From Research to Production: Lessons in ML Systems — NeurIPS Workshop (2023)
1. Efficient Deep Learning: A Practitioner's Perspective — Google Tech Talk (2022)